

Project Guidelines

General Requirements

- Team Composition: Exactly 3 students per project
- Working forensic investigation tool in GitHub repository
- Complete technical documentation and investigation guide
- Test cases with real forensic evidence samples
- Forensic validation (accuracy testing with ground truth)
- 10-15 minute presentation with live investigation demo + 5 minute Q&A

Marks Distribution (20 marks per project)

Component	Marks
Tool Functionality & Investigation Accuracy	8
Code Quality, Chain of Custody, Documentation	4
Presentation & Live Investigation Demo	5
Q&A & Forensic Knowledge	3
Total	20

Summary of All Projects

Project	Title	Focus Area	Primary Tools
1	Image Metadata & Geolocation	EXIF extraction, GPS mapping	Python, Pillow, exifread, folium
2	Social Media Forensics & OSINT	Content archiving, network graphs	Selenium, networkx, matplotlib
3	Memory Forensics	Active malware, process injection	Volatility3, YARA, Python
4	Deepfake & Image Detection	AI manipulation detection	TensorFlow, OpenCV, FaceNet
5	Quantum-Safe Evidence *	Future-proof encryption	liboqs, ML-KEM/DSA
6	AI Timeline Correlation *	Anomaly detection	TensorFlow, scikit-learn

7	Blockchain Evidence Cert *	Immutable chain of custody	Python, Ganache, Web3.py
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* = Specialized projects areas (Quantum, Blockchain, AI/ML) – Bonus

Assessment & Submission Requirements

GitHub Repository Requirements

- Public repository with clear project structure
- Comprehensive README.md with setup and usage instructions
- requirements.txt or equivalent for dependencies

Presentation Specifications

- Duration: 10-15 minutes presentation + 5 minutes Q&A
- Slide deck covering project overview, implementation details, and results
- Live demonstration of working code
- Discussion of challenges faced and solutions
- Future improvements and recommendations

Submission Deadline

- Final code and documentation: 1 week before presentation date
- GitHub link must be submitted via LMS

Project 1: Digital Image Metadata Extraction & Geolocation Analysis

Market Problem

Digital images contain hidden metadata (EXIF data) revealing location, camera model, timestamp, and even GPS coordinates. Investigators need tools to extract, visualize, and correlate geolocation data from images to track suspects, identify crime scenes, and build location timelines.

Proposed Solution

Forensic tool that: extracts EXIF metadata from images, decodes GPS coordinates, maps locations on interactive map, creates geolocation timeline, detects metadata stripping/manipulation, correlates multiple images by time/location, and generates forensic reports.

Free Tools & Technologies

- Python with Pillow for image processing
- exifread for EXIF extraction
- folium for geolocation mapping
- PyQt5 for desktop GUI

Key Requirements

1. EXIF Data Extraction (all image formats)
2. GPS Coordinate Decoding & Verification
3. Interactive Map Visualization with Timeline
4. Metadata Anomaly Detection (manipulation signs)
5. Forensic Report Generation with Evidence Chain

Project 2: Social Media Forensics & OSINT Investigation

Market Problem

Social media contains massive forensic evidence: deleted posts, account connections, location check-ins, and communication history. Investigators need tools to archive social media content, detect relationships between accounts, and create network graphs showing suspect connections.

Proposed Solution

OSINT framework that: archives social media posts/profiles, extracts account metadata, creates network graphs showing relationships, performs username search across platforms, detects deleted/hidden content through caches, and generates social network analysis reports.

Free Tools & Technologies

- Selenium for web scraping & archiving
- networkx for social graph analysis
- requests library for API access
- matplotlib for network visualization

Key Requirements

1. Social Media Content Archiving (snapshots with timestamps)
2. User Profile Extraction (bios, connections, location history)
3. Network Graph Generation (suspect relationship mapping)
4. Username Cross-Platform Search
5. Deleted Content Recovery via Cache Analysis

Project 3: Memory Forensics & Volatile Data Analysis

Market Problem

Critical evidence exists only in RAM (running processes, encryption keys, network connections). Memory dumps must be captured and analyzed to detect active malware, recover deleted data, and identify attacker artifacts before the system powers down.

Proposed Solution

Memory forensics framework that: parses memory dumps (Windows minidumps, full memory images), extracts running process trees, identifies injected code, recovers network sockets (active connections), finds encryption keys in memory, and detects malware signatures.

Free Tools & Technologies

- volatility3 memory analysis framework
- Python for custom memory parsing
- YARA rules for malware detection

- PyQt5 for interactive memory viewer

Key Requirements

1. Memory Dump Parser (Windows, Linux, Mac)
2. Process Extraction & Analysis (threads, DLLs, code caves)
3. Network Socket Recovery & Connection Tracking
4. Encryption Key & Credential Detection
5. YARA Signature Scanning for Known Malware

Project 4: AI-Powered Deepfake & Image Manipulation Detection

Market Problem

Deepfakes and manipulated images are used in fraud, blackmail, and misinformation. Investigators need AI tools to detect: AI-generated faces, doctored images, spliced content, and identify the manipulation method to establish evidence authenticity.

Proposed Solution

AI forensics tool that: detects deepfakes using facial analysis, identifies image splicing/blending, analyzes compression artifacts, detects AI-generated content patterns, performs reverse image search, and generates manipulation probability reports.

Free Tools & Technologies

- TensorFlow/OpenCV for computer vision
- FaceNet for facial recognition
- scikit-image for manipulation detection
- SHAP for AI explainability

Key Requirements

1. Deepfake Detection (facial inconsistencies, eye blinking)
2. Image Splicing Detection (boundaries, lighting analysis)
3. AI-Generated Content Detection (statistical patterns)
4. Compression Artifact Analysis
5. Manipulation Probability Scoring & Evidence Reports

Project 5: Quantum-Safe Metadata Evidence Encryption & Chain of Custody

Market Problem

Forensic metadata (location data, timestamps, communications) must remain confidential for decades. Future quantum computers could decrypt evidence stored today. Investigators need quantum-resistant encryption for evidence storage and chain of custody records.

Proposed Solution

Evidence management system with: hybrid RSA + ML-KEM encryption for metadata, ML-DSA signatures for chain of custody, tamper detection, immutable access logs, and quantum-resistant sealing of forensic evidence.

Free Tools & Technologies

- liboqs for post-quantum cryptography
- cryptography library for RSA
- PyQt5 for evidence management GUI
- SQLite for tamper-evident logs

Key Requirements

1. Hybrid Encryption (RSA + ML-KEM for evidence)
2. Chain of Custody Management (ML-DSA signatures)
3. Tamper Detection & Integrity Verification
4. Access Control & Immutable Audit Trails
5. Legal Admissibility Certificates

Project 6: AI-Powered Timeline Correlation & Anomaly Detection

Market Problem

Investigators collect data from multiple sources (logs, messages, geolocation, social media) with thousands of events. Manual timeline creation takes weeks. Need: AI to automatically correlate events, detect anomalies, and build suspect behavioral profiles.

Proposed Solution

AI forensics platform that: ingests multi-source evidence (logs, messages, location data, communications), automatically correlates events by time/actor, detects behavioral anomalies (deviations from normal patterns), generates investigator suggestions, and creates visual timelines.

Free Tools & Technologies

- scikit-learn for unsupervised anomaly detection
- TensorFlow for sequence modeling
- pandas for data aggregation
- Matplotlib/Plotly for timeline visualization

Key Requirements

1. Multi-Source Data Ingestion (logs, messages, location, comms)
2. Automatic Event Correlation & Timeline Generation
3. Behavioral Anomaly Detection (unusual patterns)
4. Actor Relationship Network Building
5. Interactive Timeline Visualization with Alerts

Project 7: Blockchain Metadata Chain of Custody & Evidence Verification

Market Problem

Forensic metadata (location data, timestamps, relationships) can be disputed in court. Investigators need immutable, third-party verifiable proof that evidence hasn't been tampered with. Blockchain provides cryptographic guarantees of evidence integrity.

Proposed Solution

Blockchain evidence verification system that: records evidence hashes on blockchain, logs chain of custody transactions, timestamps each access, allows third-party verification, generates legally-admissible blockchain certificates proving evidence integrity.

Free Tools & Technologies

- Python for blockchain implementation
- Ganache for local blockchain nodes
- Web3.py for smart contracts
- SQLite for case metadata

Key Requirements

1. Blockchain Core (SHA-256, chain validation)
2. Evidence Hash Registration (immutable records)
3. Chain of Custody Transaction Logging
4. Smart Contracts for Access Control
5. Third-Party Verification & Legal Certificates